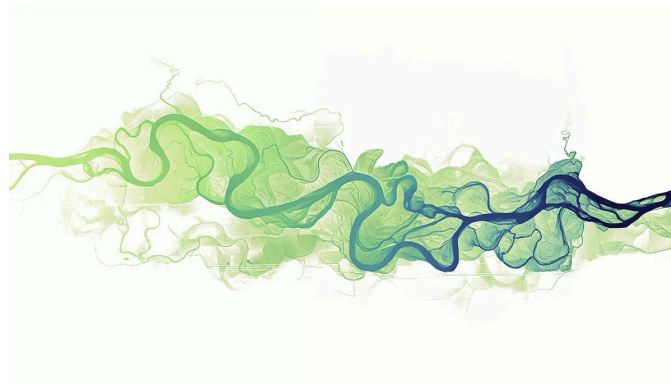


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Advanced Earth Observation for Grassland Management:

Improving Automated Mowing Detection Using Multimodal Remote Sensing and Machine Learning



Bachelor's Thesis

by
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Abstract

1. Introcutiion

Grassland management at airports is a safety-critical task. Vegetation height and structure on airfield grounds directly influence habitat attractiveness for bird species, which in turn affects the frequency and severity of bird strikes (DeVault et al. 2011). At Zurich Airport, grasslands are subject to regular mowing protocols designed to reduce wildlife hazards. However, monitoring when and where mowing occurs has traditionally relied on manual records created by field staff, a process that is labour-intensive and prone to spatial inaccuracies. To investigate the relationship between mowing practices and bird population dynamics, consistent and spatially reliable mowing records are required.

Satellite-based remote sensing offers a scalable alternative to manual monitoring. The Sentinel-2 constellation, operated by the European Space Agency, provides multispectral imagery at 10 to 20 metre spatial resolution with a revisit cycle of approximately five days (Drusch et al. 2012). Vegetation indices derived from these data, such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Infrared Index (NDII), capture the spectral response of vegetation to management events like mowing, where the abrupt removal of biomass produces a measurable drop in reflectance values (Pettorelli et al. 2005; Andreatta et al. 2022).

In a preceding project work, a pixel-based machine learning workflow for automated mowing detection at Zurich Airport using Sentinel-2 imagery was developed (Klaver 2025). That study demonstrated that standard classifiers, particularly the Support Vector Machine, can detect mowing events with high accuracy on controlled test samples (F1-Score of 0.923). However, the work also identified two fundamental limitations that prevent the system from functioning as a reliable operational tool.

The first limitation concerns data availability. Optical sensors depend on clear atmospheric conditions, and in the temperate climate of central Switzerland, frequent cloud cover regularly interrupts the temporal chain of observations needed for change detection. This problem is well documented in the literature. De Vroey et al. (2021) highlight that relying solely on optical data is particularly problematic in regions where cloud cover is persistent during the growing season. Synthetic Aperture Radar (SAR), which operates independently of weather and illumination conditions, has been proposed as a complementary data source. Several studies have demonstrated that interferometric coherence derived from SAR acquisitions can detect vegetation changes such as mowing, because the removal of biomass alters the scattering properties of the surface (Tamm et al. 2016; Zalite et al. 2016; Reinermann et al. 2022).

The second limitation concerns spatial consistency. The pixel-based classification approach produces predictions independently for each pixel, resulting in spatially fragmented output with characteristic salt-and-pepper noise (Blaschke 2010). In the full-scene application conducted in the previous study, models that performed well on test metrics produced scattered false positives across the scene and failed to reconstruct the contiguous shapes of mowing events (Klaver 2025). Spatial postprocessing techniques, such as morphological filtering (Jamil et al. 2008) and superpixel-based segmentation (Achanta et al. 2012), can address this issue by incorporating neighbourhood context into the classification output. These methods have been applied in remote sensing contexts to improve the spatial coherence of pixel-based classifications (Dharani and Sreenivasulu 2021; Crommelinck et al. 2017), though their specific application to mowing detection remains underexplored.

The main objectives are:

1. To investigate the integration of Synthetic Aperture Radar (SAR) data from the Sentinel-1 mission as a complementary input to the existing optical feature set, with the goal of reducing dependence on cloud-free optical acquisitions.
2. To implement and evaluate spatial postprocessing methods that improve the spatial consistency of pixel-based predictions.
3. To benchmark the enhanced workflow against the existing baseline model from the previous study, assessing whether the additions yield measurable improvements in detection robustness and operational reliability.

2. Literature Review

This chapter reviews the theoretical foundations relevant to the two main methodological extensions introduced in this thesis: the integration of Synthetic Aperture Radar (SAR) data and the application of spatial postprocessing techniques. For a detailed treatment of optical remote sensing with Sentinel-2, vegetation index theory, and the pixel-based machine learning workflow that serves as the baseline for this work, please refer to the preceding project work (Klaver 2025).

2.1. Optical Remote Sensing for Grassland Monitoring

Satellite-based optical remote sensing enables continuous observation of vegetation dynamics across large spatial and temporal scales. The Sentinel-2 mission provides multispectral imagery in 13 spectral bands spanning visible, near-infrared, and short-wave infrared wavelengths, with spatial resolutions between 10 and 60 metres and a revisit cycle of approximately five days (Drusch et al. 2012). Vegetation indices derived from these bands quantify biophysical properties of vegetation. The Normalized Difference Vegetation Index (NDVI), which exploits the contrast between red absorption and near-infrared reflectance, is the most widely used indicator of vegetation health and biomass (Pettorelli et al. 2005). The Normalized Difference Infrared Index (NDII), which incorporates the short-wave infrared band sensitive to leaf water content, has been shown to yield high accuracy for mowing frequency detection, in some cases outperforming NDVI-based approaches (Andreatta et al. 2022).

Multiple studies have investigated automated mowing detection from optical time series using both rule-based and machine learning approaches. Reinermann et al. (2022) applied a threshold-based algorithm to Sentinel-2 EVI time series in Germany. Machine learning classifiers such as Random Forest and Support Vector Machines have also been employed, with Garioud et al. (2019) noting that simpler models can be highly effective when training data is limited. A comprehensive review of remote sensing approaches for grassland production and management is provided by Reinermann et al. (2020).

Despite the effectiveness of optical approaches, their operational reliability is constrained by atmospheric conditions. In temperate climates, cloud cover frequently disrupts the temporal continuity of observations required for change detection (De Vroey et al. 2021). This limitation motivates the integration of weather-independent data sources such as SAR.

2.2. Synthetic Aperture Radar

Synthetic Aperture Radar (SAR) is an active microwave imaging technology that differs fundamentally from optical sensors. While optical systems passively record reflected sunlight, a SAR instrument transmits its own microwave pulses and records the backscattered signal. Because SAR operates at microwave wavelengths, it can penetrate cloud cover and acquire data independently of solar illumination (Meyer 2019). As illustrated in Figure 1, SAR also differs from optical systems in its observation geometry: the antenna is pointed at an oblique look angle rather than towards nadir, and the image is formed by sorting returned echoes according to their arrival time and Doppler shift. The term “synthetic aperture” refers to the technique of combining echoes from multiple positions along the flight path to simulate a much longer antenna, enabling metre-scale resolution from orbital altitudes (Meyer 2019).

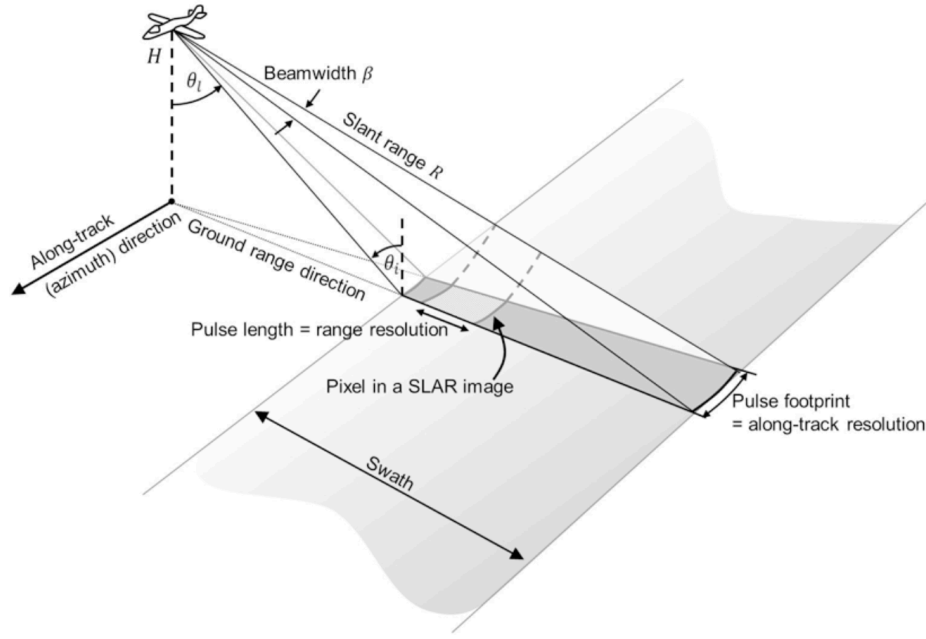


Figure 1: Observation geometry of a side-looking SAR system. The radar sensor transmits pulses at an oblique look angle and records the backscattered signal. Adapted from Meyer (2019).

A SAR sensor records both the amplitude and the phase of the returned signal. The amplitude depends on surface properties such as roughness, moisture content, and vegetation structure. The phase encodes the round-trip distance between sensor and surface and is essential for interferometric techniques discussed below. SAR images are additionally characterised by a granular appearance known as speckle, an inherent property of coherent imaging systems that can be reduced through spatial averaging at the cost of resolution (Meyer 2019).

The Sentinel-1 Mission

The Sentinel-1 mission operates a C-band SAR instrument at a wavelength of approximately 5.6 cm. In its primary Interferometric Wide Swath (IW) mode, it achieves a spatial resolution of 5 by 20 metres with a 250 km swath width (ESA 2013). The mission distributes data in different product types: Level-1 Single Look Complex (SLC) products retain both amplitude and phase information, while Level-1 Ground Range Detected (GRD) products discard the phase during processing. Consequently, only SLC products can be used for interferometric analysis (ESA 2013; Copernicus Data Space Ecosystem 2025).

Interferometric Coherence

When two SAR acquisitions are taken from nearly the same orbital position at different times, their complex signals can be compared through a technique known as SAR interferometry (InSAR). The interferometric coherence quantifies the degree of similarity between the two complex signals and is expressed as a value between 0 and 1 (Tamm et al. 2016). Given two complex SAR images s_1 and s_2 , the coherence γ is calculated as:

$$\gamma = \frac{|\langle s_1 s_2^* \rangle|}{\sqrt{\langle s_1 s_1^* \rangle \langle s_2 s_2^* \rangle}}, \quad 0 \leq \gamma \leq 1 \quad (1)$$

where $|\cdot|$ denotes the absolute value, $\langle \cdot \rangle$ denotes a spatial averaging operation and $*$ denotes the complex conjugate. A coherence value near 1 indicates that the position and scattering properties

of the surface elements remained stable between the two acquisitions, while a value near 0 indicates complete decorrelation (Tamm et al. 2016).

For vegetated surfaces, temporal decorrelation is driven by continuous changes in canopy geometry through leaf movement, growth, and moisture fluctuations, which cause progressive phase decorrelation. When vegetation is removed through mowing, the exposed soil or short stubble presents a more stable scattering surface, leading to higher coherence. This mechanism makes interferometric coherence a potentially useful indicator for mowing detection.

The measured coherence is not solely determined by temporal change on the ground. Tamm et al. (2016) decompose the total estimated coherence as $\gamma_{total} = \gamma_{temporal} \cdot \gamma_{SNR} \cdot \gamma_{bias} \cdot \gamma_{other}$, where $\gamma_{temporal}$ captures physical surface changes and, γ_{SNR} accounts for decorrelation due to system thermal noise, and the remaining terms capture processing-related and volumetric effects. For most applications the non-temporal terms are negligible, but the weak interaction between C-band radar and grassland vegetation produces lower signal-to-noise ratios, making SNR correction non-negligible for grassland monitoring (Tamm et al. 2016).

The sensitivity of SAR to vegetation also depends on radar wavelength. As shown in Figure 2, shorter wavelengths such as X-band (approximately 3 cm) interact primarily with the upper canopy, while C-band (approximately 6 cm) and L-band (approximately 23 cm) penetrate deeper into the vegetation (Meyer 2019). This distinction is relevant when comparing studies: Zalite et al. (2016) confirmed that X-band interferometric coherence is sensitive to mowing, but coherence responses may differ in magnitude from C-band due to the different penetration depths.

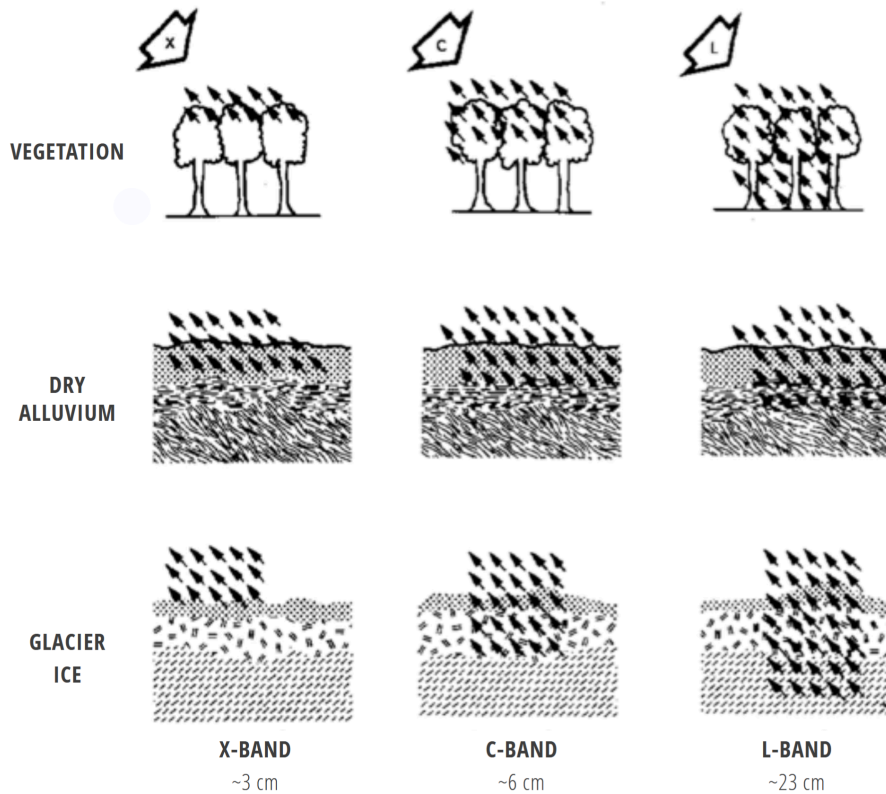


Figure 2: SAR signal penetration by sensor wavelength λ . Adapted from Meyer (2019)

Tamm et al. (2016) investigated the coherence response to mowing using Sentinel-1 C-band data with 12-day temporal baselines. Both VH and VV polarisation coherence values were significantly higher after mowing, with the strongest increase occurring shortly after the event and elevated

values persisting for up to 36 days. However, precipitation decreased coherence regardless of vegetation state, and rapid regrowth within the 12-day revisit period reduced the coherence contrast.

SAR-Optical Fusion for Mowing Detection

Several studies have combined SAR and optical data for mowing detection with mixed results. De Vroey et al. (2021) found that 54% of mowing events in hay meadows could be detected from Sentinel-1 coherence jumps, with grazing events identified as a major source of false detections. Reinermann et al. (2022) tested an integrated S1+S2 approach for Germany, where the optical-only method achieved an F1-Score of 0.64 but the combined approach reduced this to 0.61 due to increased false positives. Komisarenko et al. (2022) used a deep learning model with both S1 and S2 time series in Estonia, achieving an event accuracy of 73.3%. These findings highlight that SAR-optical fusion does not automatically improve performance and that careful integration strategies are necessary.

2.3. Spatial Postprocessing Methods

Pixel-based classification methods treat each pixel independently, without considering the spatial relationship between neighbouring predictions. This frequently produces fragmented classification outputs containing isolated misclassified pixels, commonly referred to as salt-and-pepper noise (Blaschke 2010). Object-Based Image Analysis (OBIA), which segments images into meaningful objects prior to classification, has been proposed as one paradigm to address this limitation (Blaschke 2010). However, OBIA requires fundamentally restructuring the classification workflow. An alternative is to retain the pixel-based approach and apply spatial postprocessing to the classification output. This preserves the simplicity of per-pixel classifiers while using spatial context to reduce noise after the fact. The following methods represent three such postprocessing strategies at different spatial scales.

A simple baseline approach is the median filter, which replaces each pixel's class label with the most frequent label within a local neighbourhood window, suppressing isolated misclassifications while largely preserving edges.

Mathematical morphology provides a more structured set of spatial operations for refining binary or categorical classification outputs. The two fundamental operations are erosion, which removes small isolated regions by shrinking object boundaries, and dilation, which expands regions and fills small gaps. Combinations of these are commonly used in practice: opening (erosion followed by dilation) removes small spurious objects, while closing (dilation followed by erosion) fills small holes and smooths boundaries (Jamil et al. 2008). These operations have been applied in remote sensing classification workflows to reduce isolated false detections while preserving the broader structure of larger classified regions (Dharani and Sreenivasulu 2021). For mowing detection, where events are expected to affect spatially contiguous areas within grassland parcels rather than isolated single pixels, morphological postprocessing can bring pixel-level predictions closer to the expected spatial scale of the target phenomenon.

An alternative approach to spatial regularisation is superpixel segmentation, which groups neighbouring pixels into compact, spectrally homogeneous clusters. The Simple Linear Iterative Clustering (SLIC) algorithm is one of the most widely used methods for this purpose. SLIC adapts k-means clustering to operate jointly in the spatial and spectral domain, producing approximately uniform segments efficiently with $O(N)$ complexity (Achanta et al. 2012). The compactness and size of the resulting segments are controlled by two parameters: the number of desired superpixels

and a compactness weight that balances spatial proximity against spectral similarity (Achanta et al. 2012). In geospatial applications, SLIC has been applied to high-resolution imagery for object delineation, where its segments have been shown to adhere well to natural image boundaries (Crommelinck et al. 2017). In a classification postprocessing context, pixel-level predictions can be aggregated within each superpixel segment through majority voting, assigning all pixels within a segment the dominant class label. This enforces local spatial consistency and reduces isolated misclassifications within otherwise homogeneous areas.

3. Methods

4. Results

5. Discussion

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Appendices

A. Raw Data

B. Statistical Outputs